



Externally bonded FRP reinforcement for RC-Structures

dr. ir. L. Vasseur

TRADECC nv

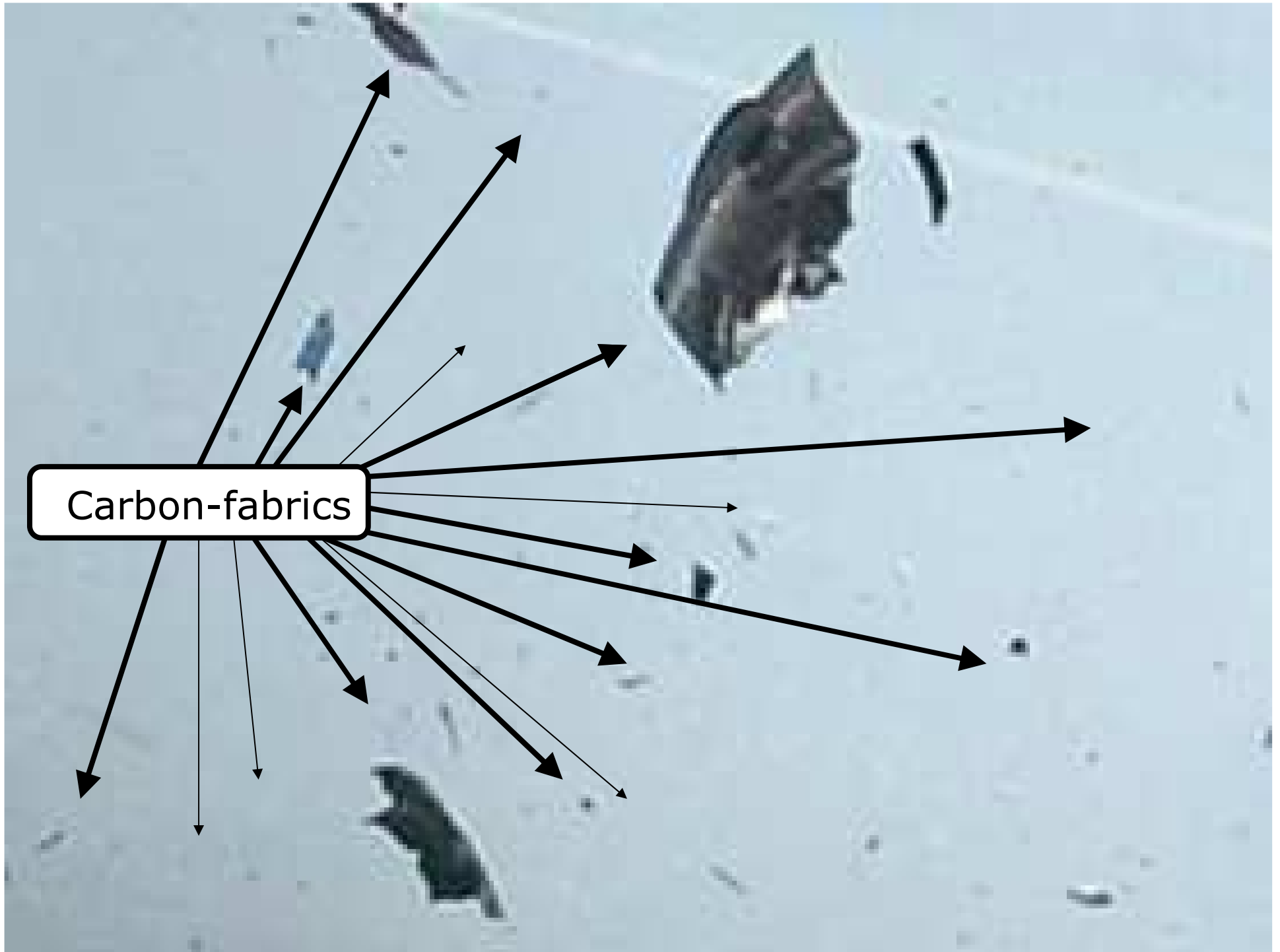
University of Johannesburg

Johannesburg

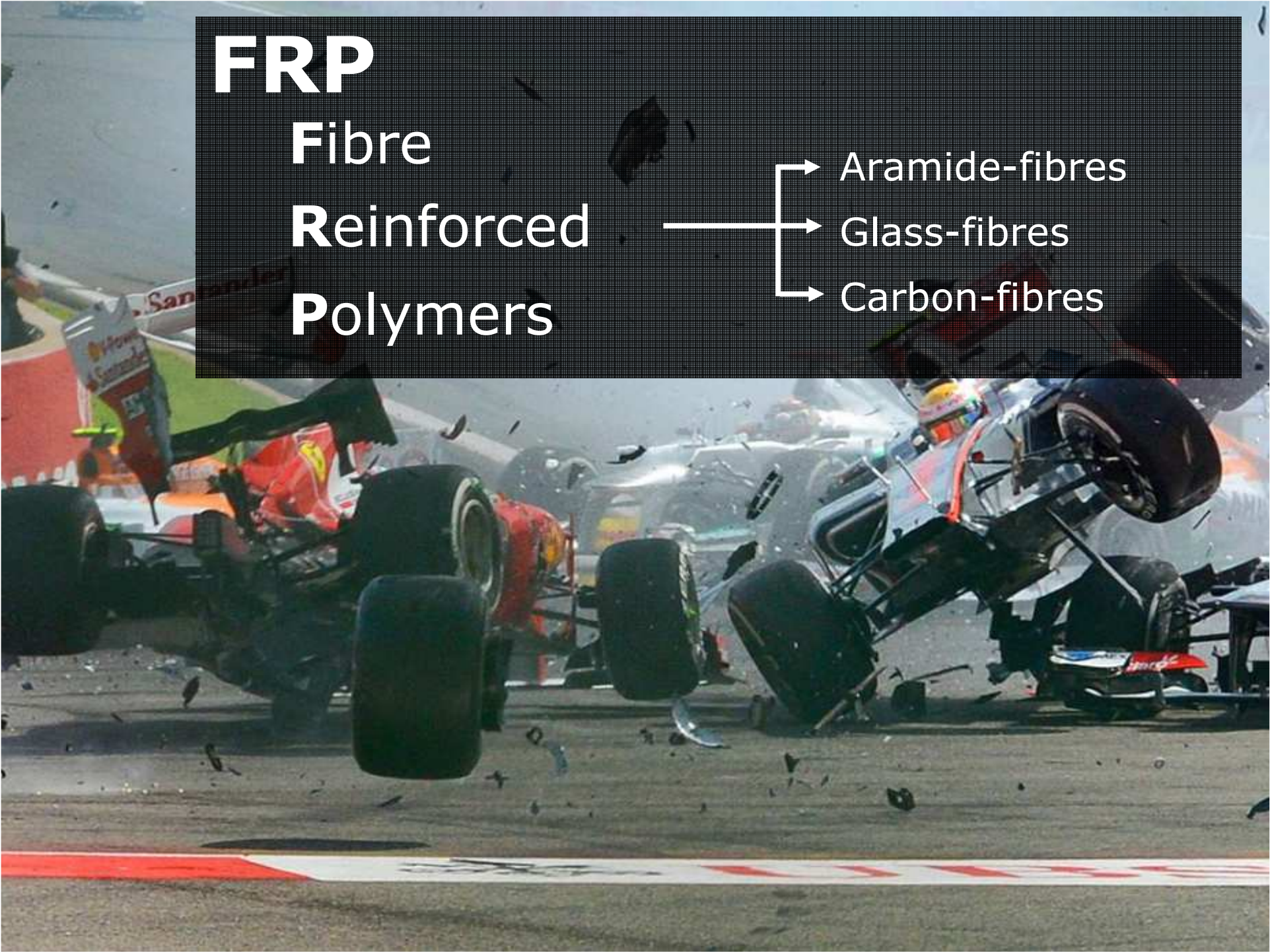
07 - 09 - 2012



Carbon-fabrics







FRP

Fibre

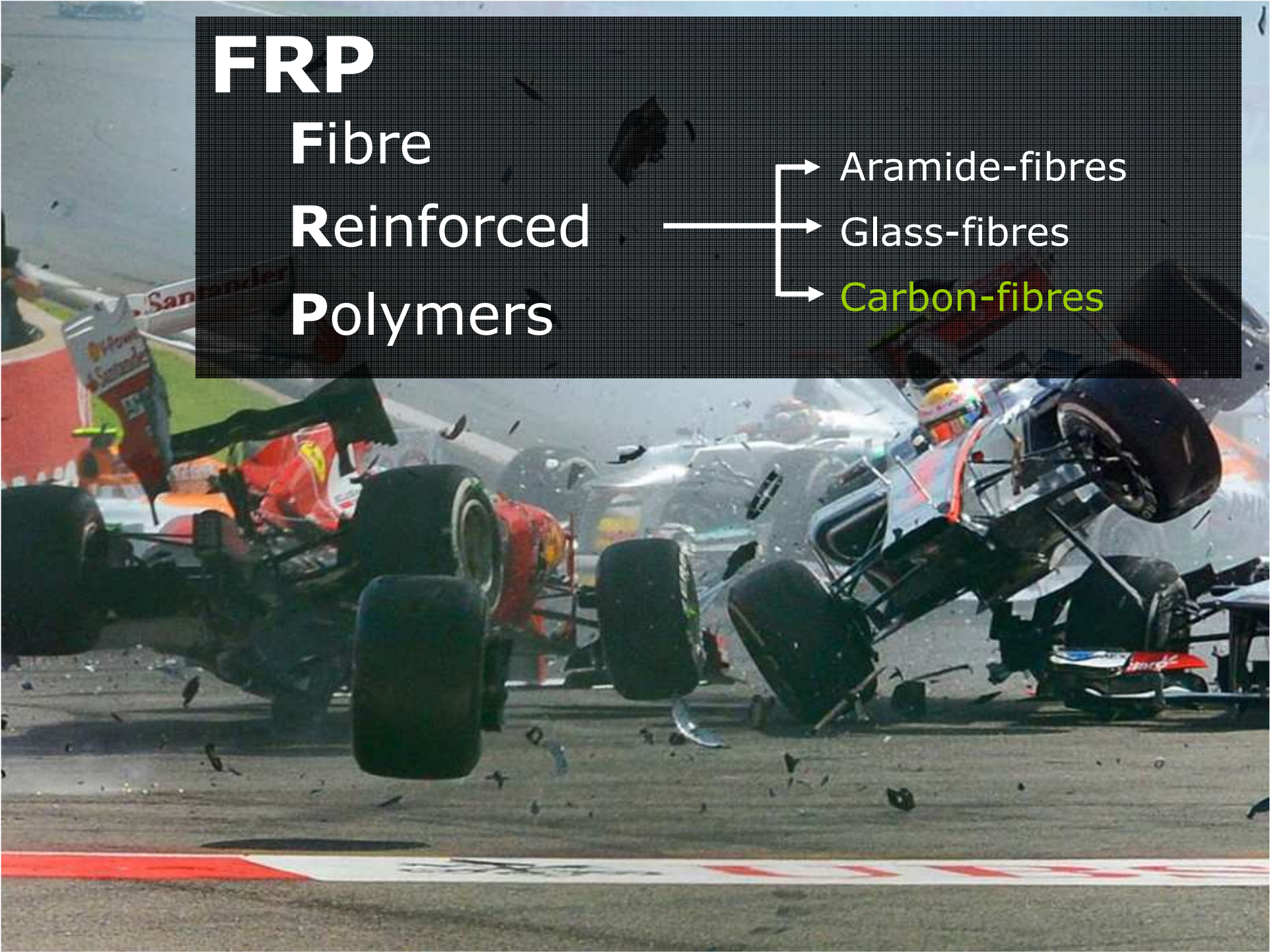
Reinforced

Polymers

→ Aramide-fibres

→ Glass-fibres

→ Carbon-fibres



FRP

Fibre

Reinforced

Polymers

→ Aramide-fibres

→ Glass-fibres

→ Carbon-fibres

FRP

Fibre

Reinforced

Polymers

applied in different fields

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

FRP

Fibre

Reinforced

Polymers



aviation

applied in different fields

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

FRP

Fibre

Reinforced

Polymers

applied in different fields



aviation



motor sport

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

FRP

Fibre

Reinforced

Polymers

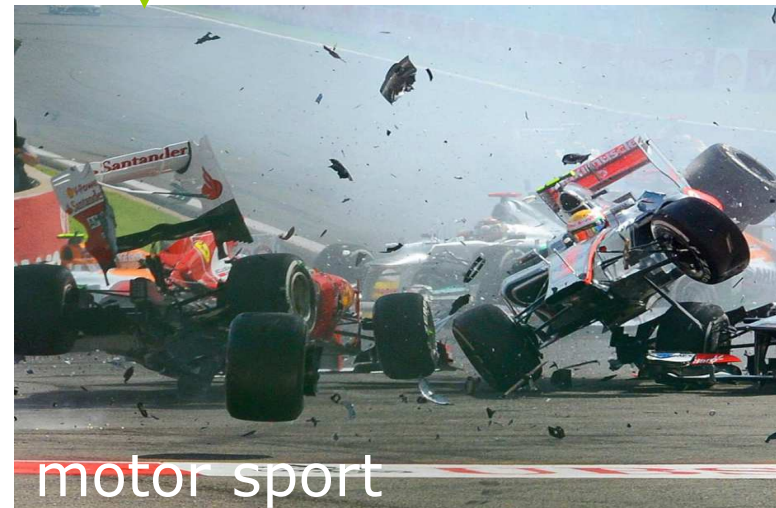
applied in different fields



aviation



civil engineering



motor sport

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

Advantages

- low weight
- no corrosion
- high strength / high E-modulus
- no loss of free height under the structure
- easy for application
- good resistance against aggressive environments
- good resistance against fatigue



TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

TRADECC

- INTRODUCTION

- TYPE OF FIBRES (CFRP – GFRP - AFRP)

- APPLICATIONS
 - Flexural Strengthening
 - Shear Strengthening
 - Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen

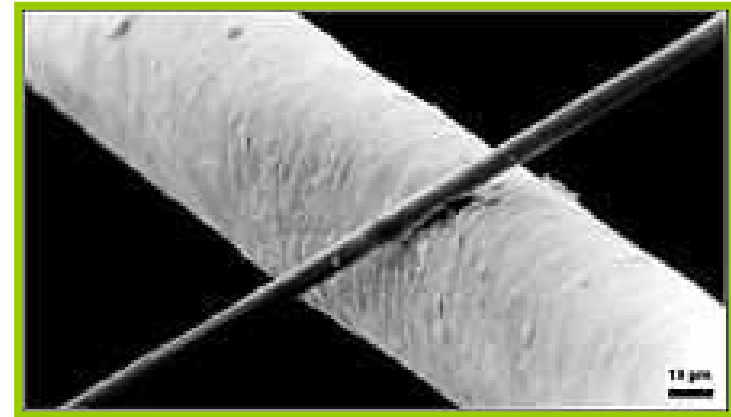


TRADECC

TYPE OF FIBRES

Carbon fibres (CFRP)

- Nylon fibres which are carbonized (increased temperatures) in an anaerobic environment
- Tensile strength: 3500 - 6000 MPa
E-modulus: 215 - 235 GPa
- Good chemical resistance and H₂O-resistance
- No influence of creep



TYPE OF FIBRES

Glas Fibres (GFRP)

- Quartz poured into a mould (at high temperatures)
- Tensile strength: 1600 - 3000 MPa
E-Modulus: 70 - 90 GPa
- No alkali resistance
- In contact with $H_2O \rightarrow$ decrease of resistance
- Creep



TYPE OF FIBRES

Glas Fibres (GFRP)

- Quartz poured into a mould (at high temperatures)
- Tensile strength: 1600 - 3000 MPa
E-Modulus: 70 - 90 GPa
- No alkali resistance
- In contact with $H_2O \rightarrow$ increase of resistance
- Creep



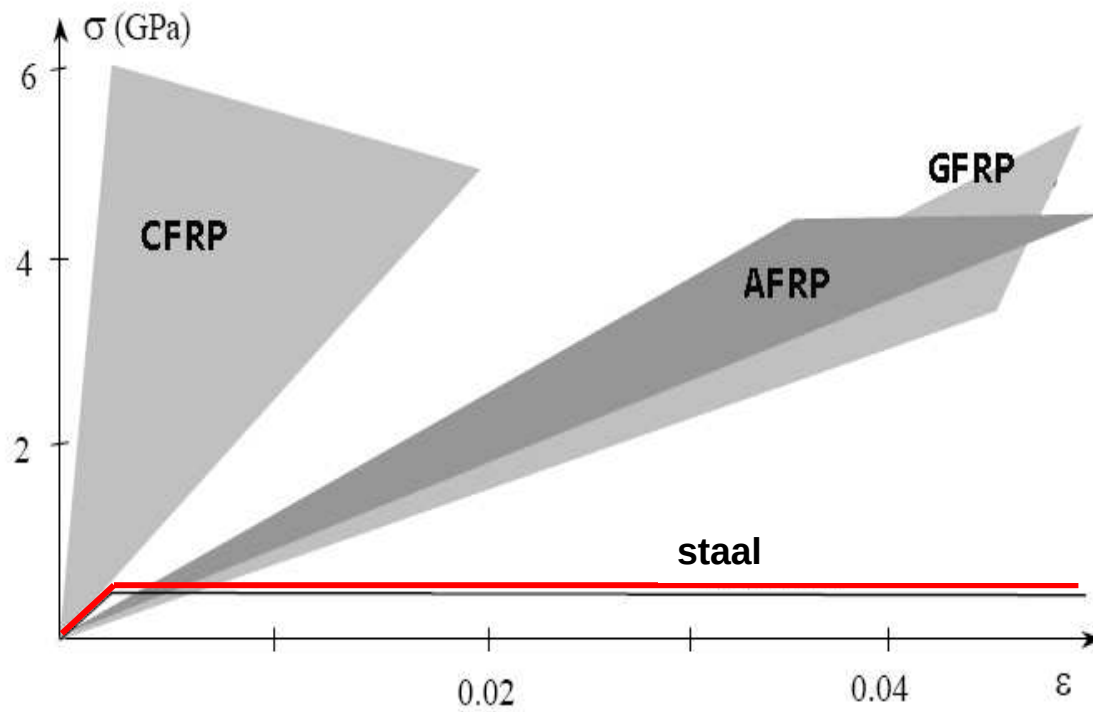
TYPE OF FIBRES

Aramide-fibres (AFRP)

- Tensile strength: 3500 - 4000 MPa
E-Modulus: 70 - 130 GPa
- Alkali resistance
- In contact with H_2O → reduction resistance
- Creep sensitive



TYPE OF FIBRES



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS

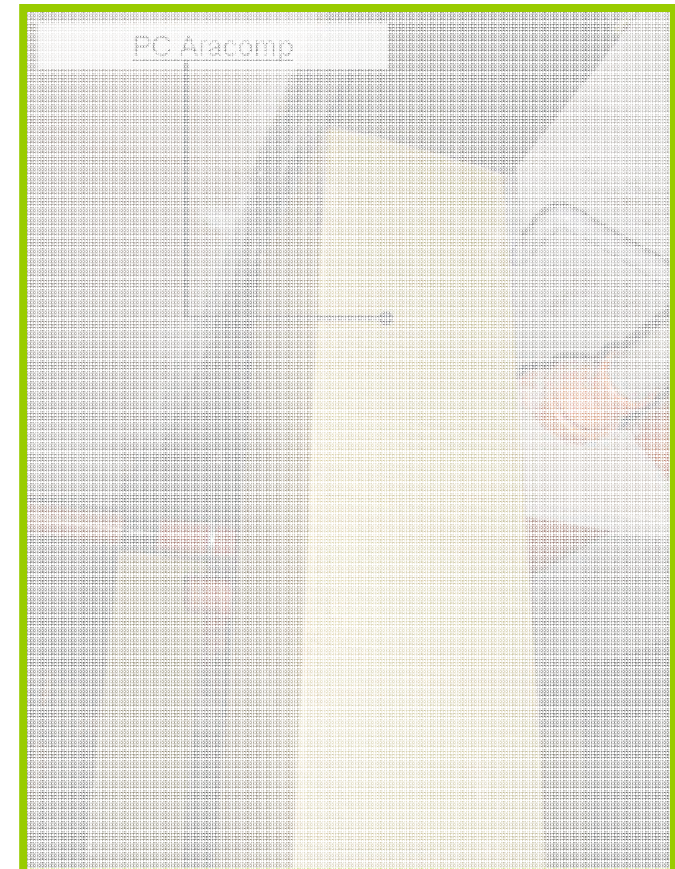


TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – Flexural reinforcement

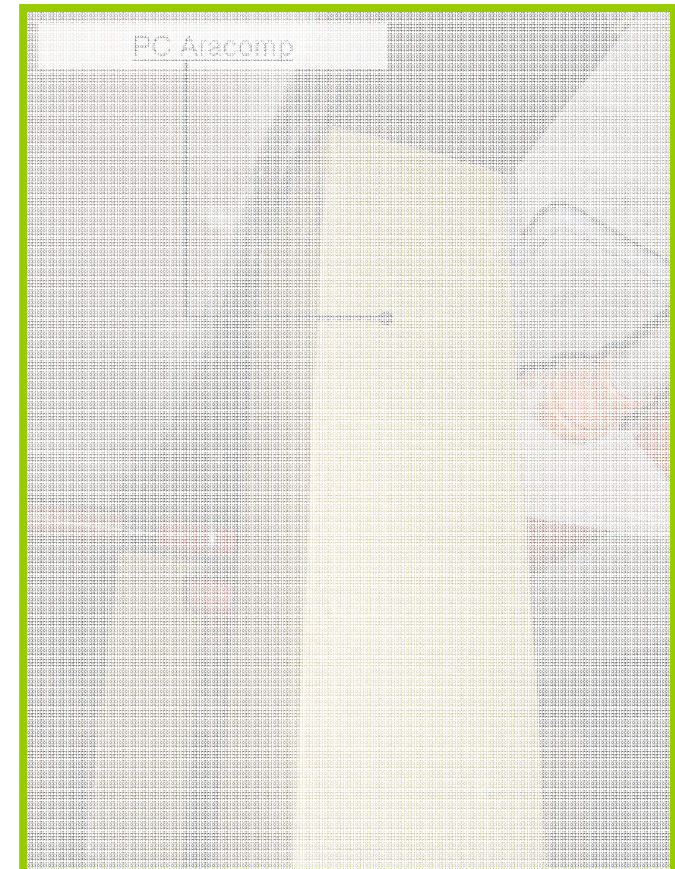


TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – Shear reinforcement

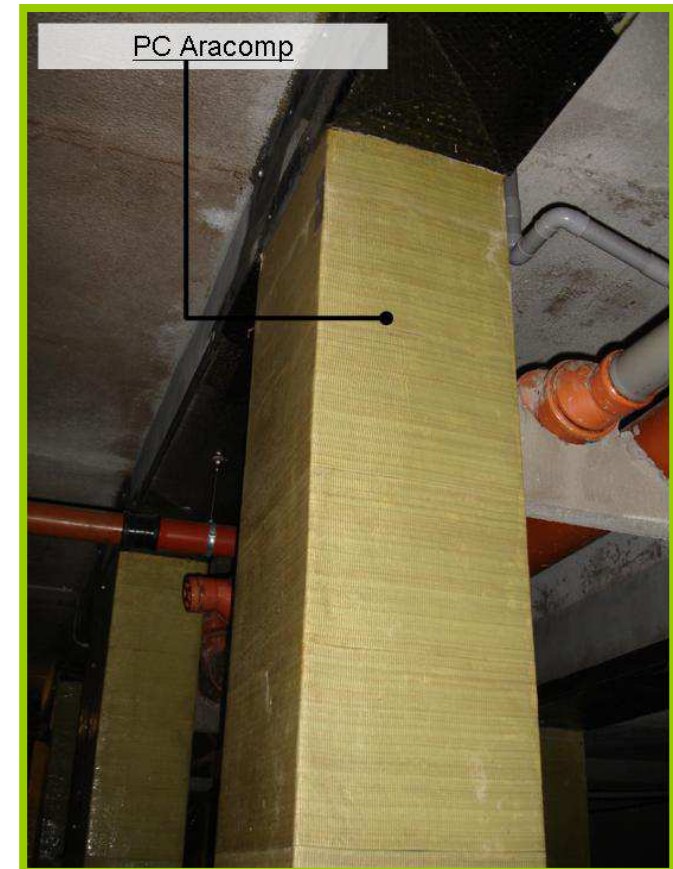
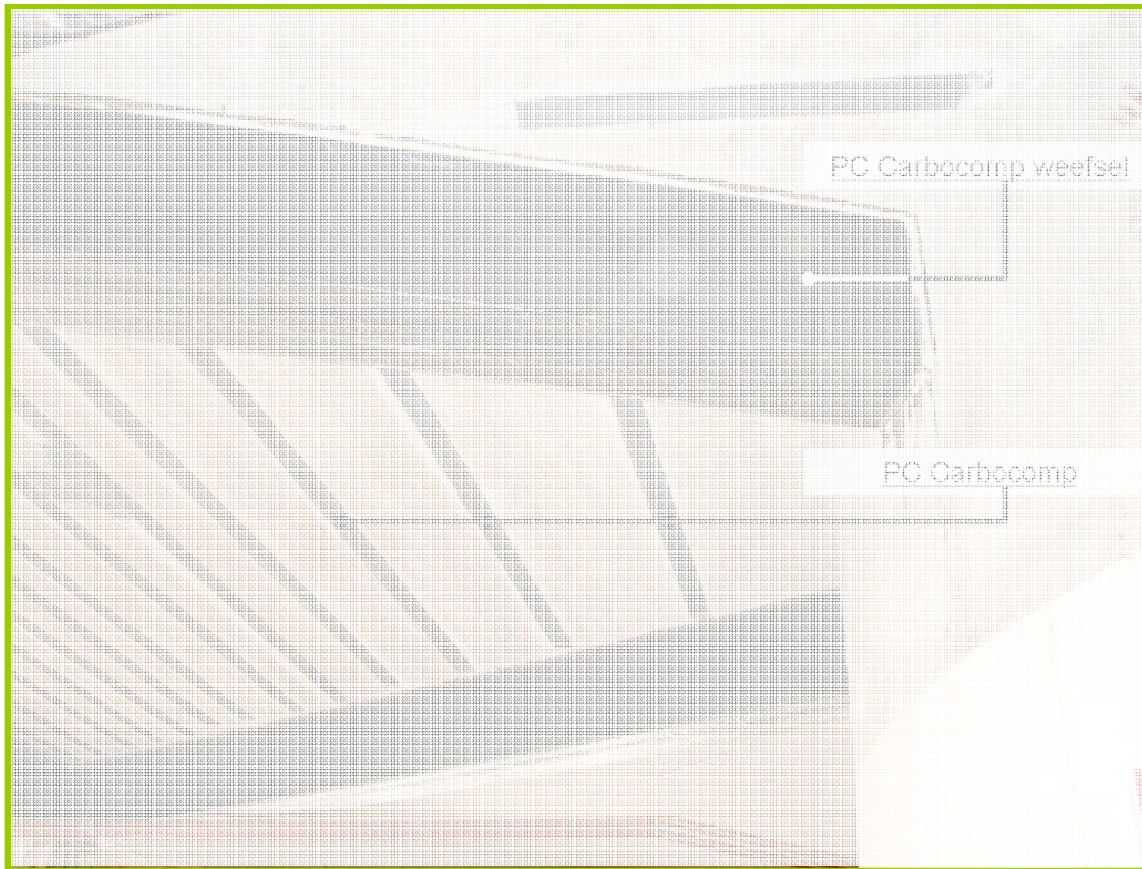


TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – Reinforcement of columns



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS

Guideline:

**FIB bulletin 14 (2001),
*Externally bonded FRP reinforcement for RC structures.***

⇒ European guideline

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



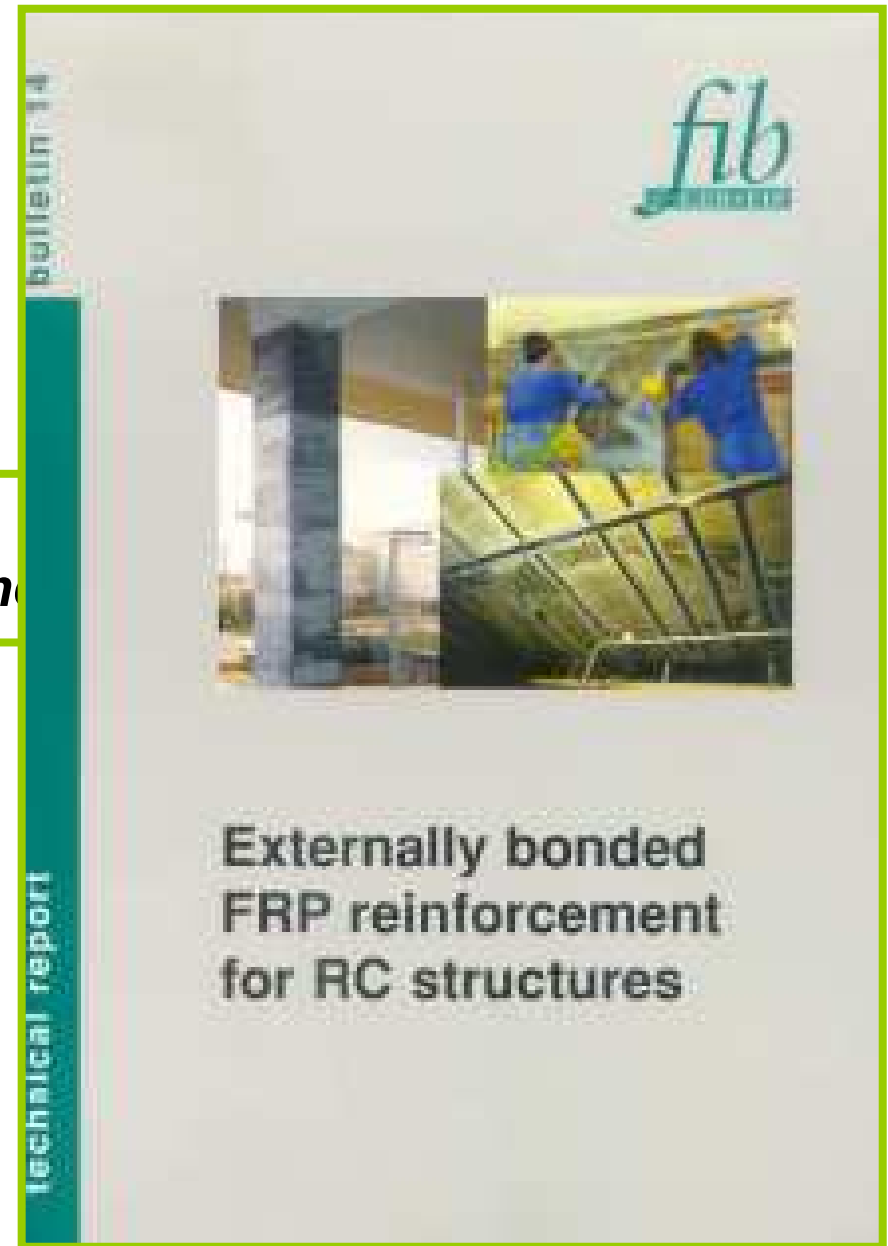
TRADECC

APPLICATIONS

Guideline:

**FIB bulletin 14 (2001),
*Externally bonded FRP reinforcement***

⇒ European guideline



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS

Guideline:

**FIB bulletin 14 (2001),
*Externally bonded FRP reinforcement for RC structures.***

Design of FRP and Steel Plated RC Structures
D. J. Oehlers and R. Seracino

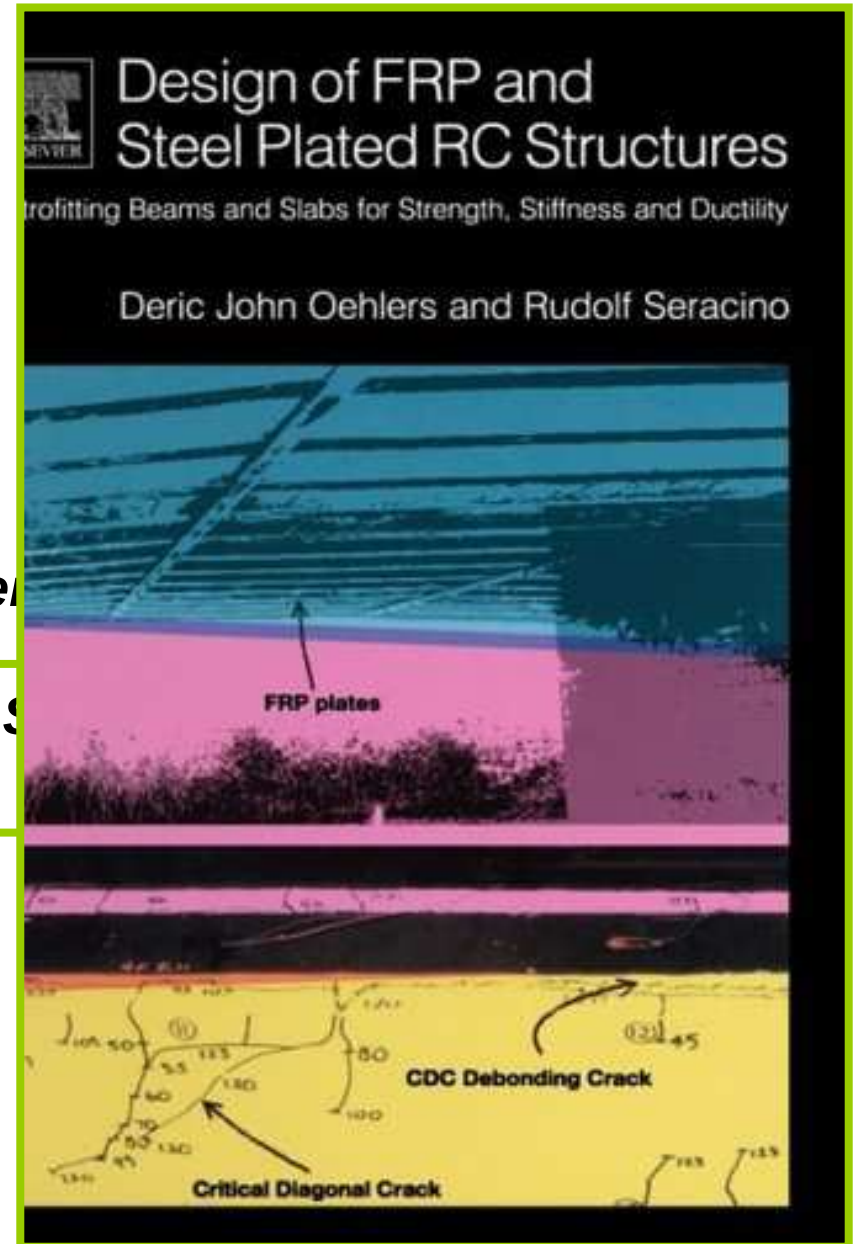


APPLICATIONS

Guideline:

FIB bulletin 14 (2001),
Externally bonded FRP reinforcement

Design of FRP and Steel Plated RC Structures
D. J. Oehlers and R. Seracino



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS

Guideline:

**FIB bulletin 14 (2001),
*Externally bonded FRP reinforcement for RC structures.***

Design of FRP and Steel Plated RC Structures
D. J. Oehlers and R. Seracino

~ conform EC2 : except taking into account:

- **Initial State**

- **Debonding Mechanisms**



APPLICATIONS – flexural strengthening

Initial state:

Loading on the structure during strengthening

Final state:

ULS: Ultimate limit state > control the **strength** of the structure

Non-linear calculation

Debonding Mechanisms

SLS: Serviceability limit state > control the **stiffness** of the structure

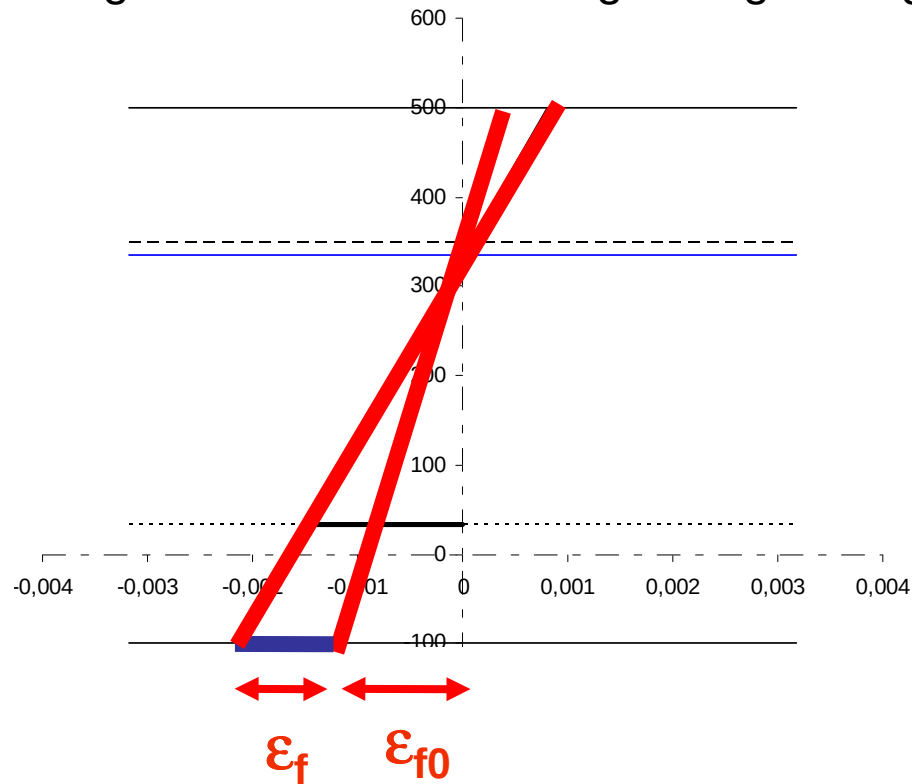
linear elastic calculation



APPLICATIONS – flexural strengthening

Initial state :

Loading on the structure during strengthening



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

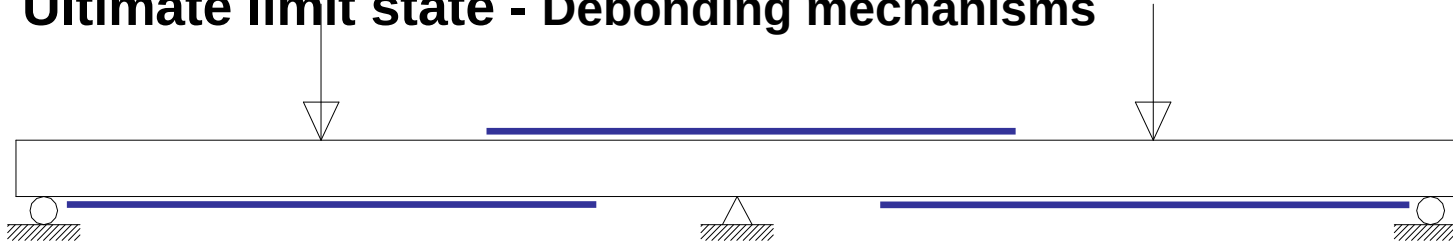
TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

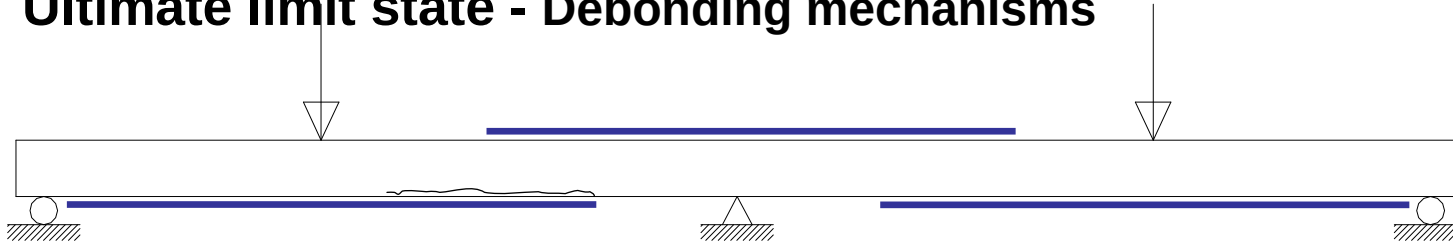
APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



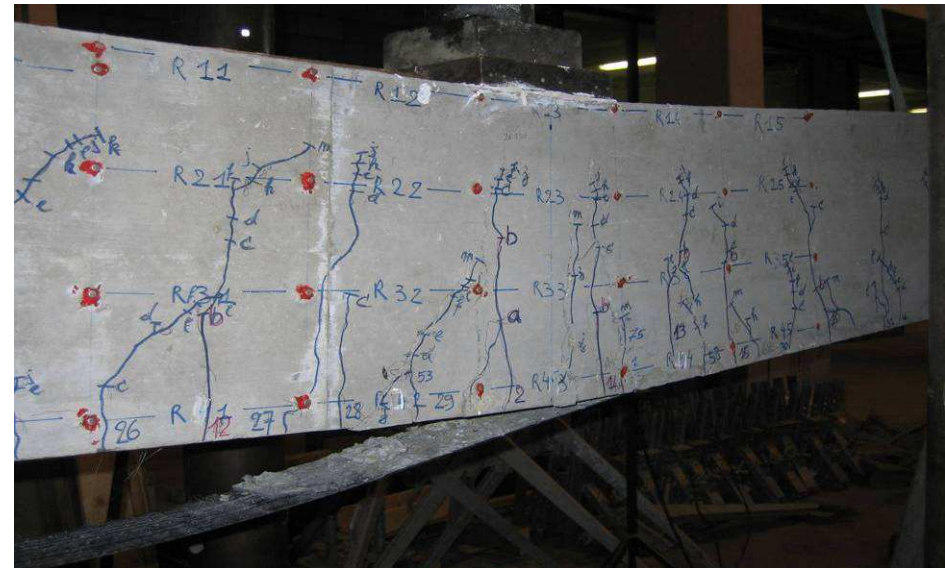
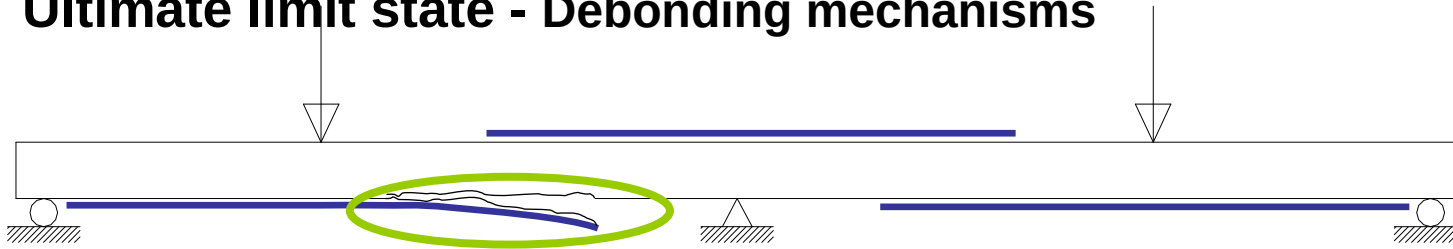
APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



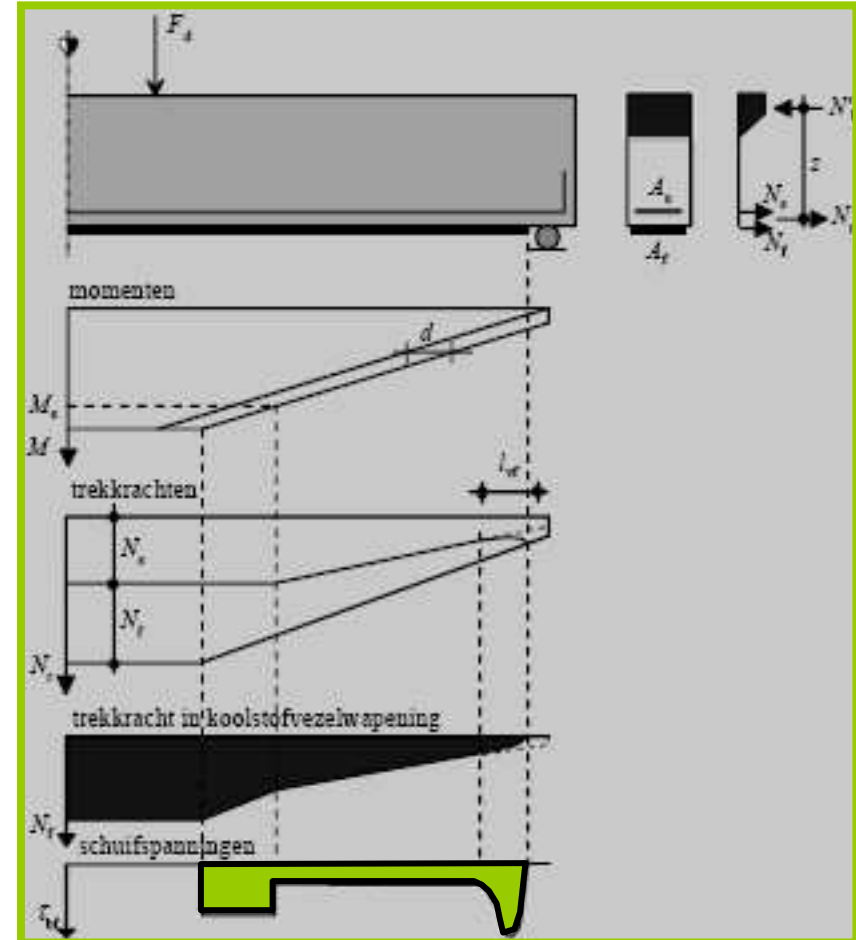
TRADECC

APPLICATIONS – flexural strengthening

Source: Doctorates thesis Stijn Matthys - UGent

ULS: Ultimate limit state

Debonding mechanisms



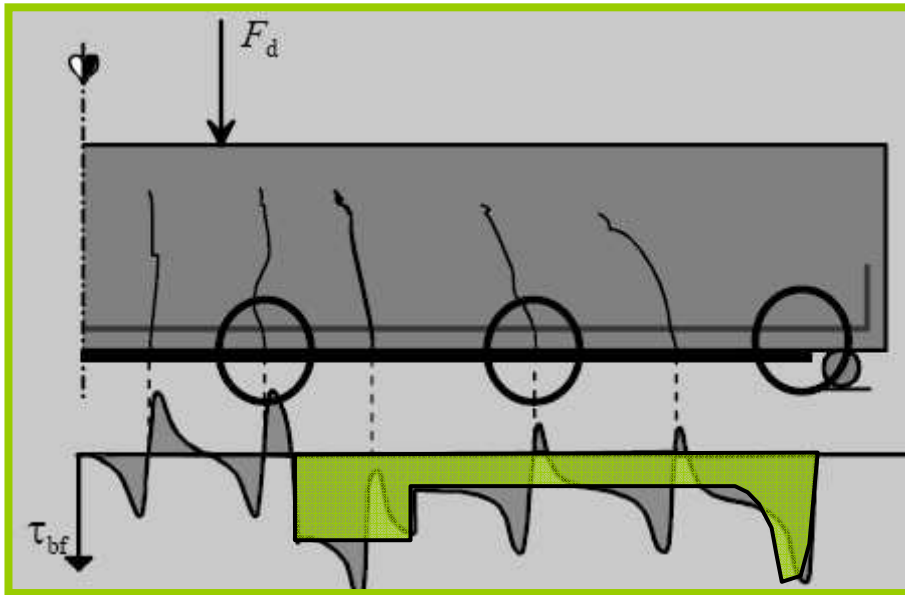
TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

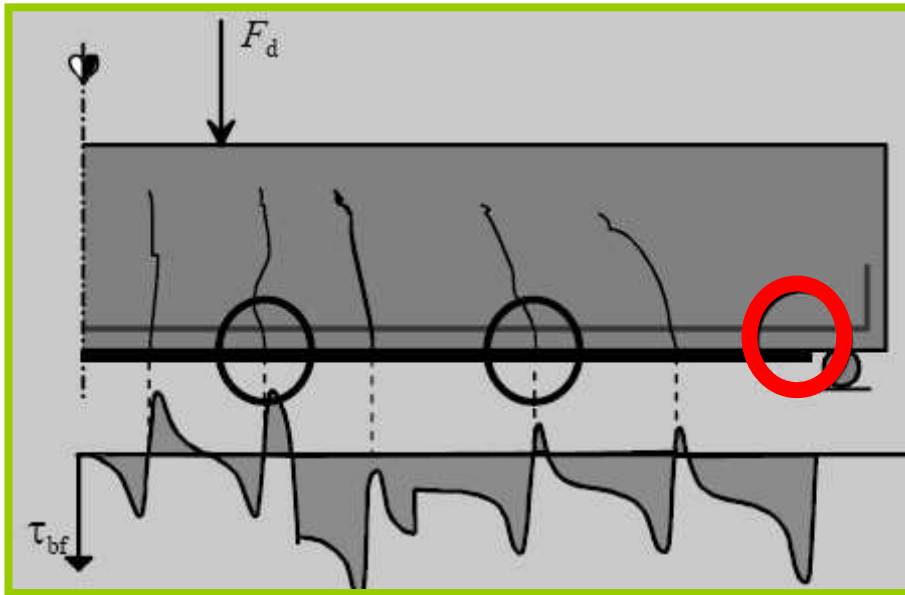


Debonding:



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



Debonding :

- End anchorage

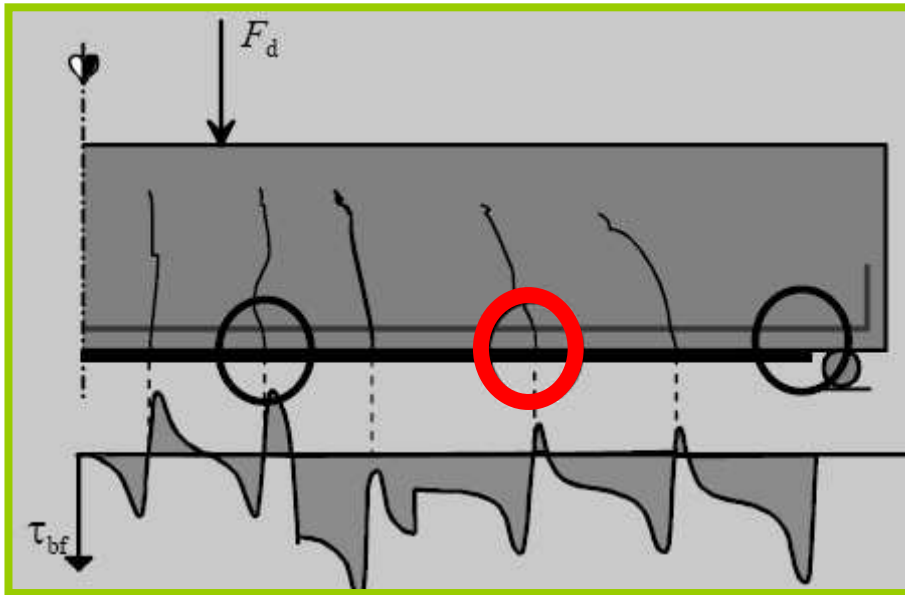
TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



Debonding :

- End anchorage
- Shear cracks

TRADECC nv

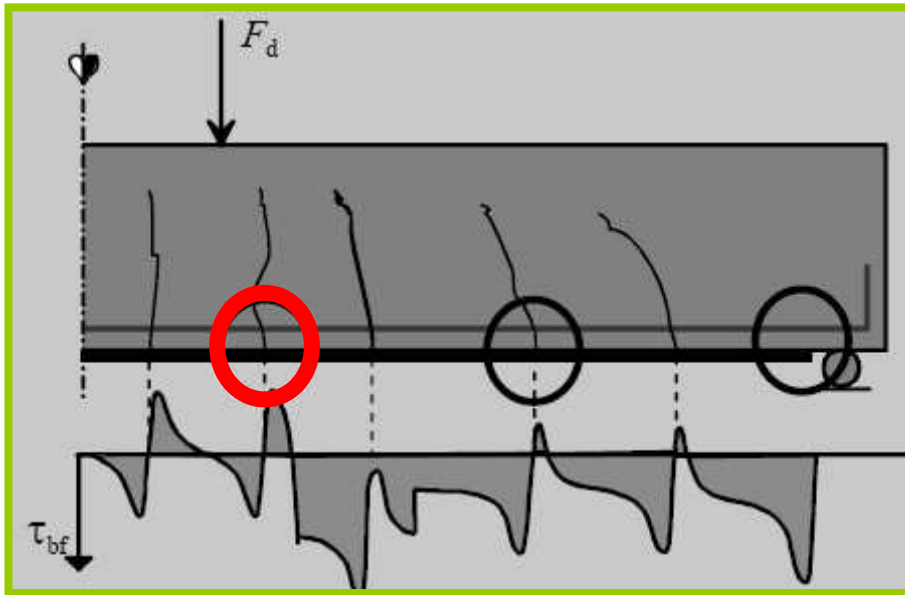
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



Debonding :

- End anchorage
- Shear cracks
- Flexural cracks



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

Following fib Bulletin

1. Anchorage failure
2. End shear failure (Concrete rip-off)
3. Debonding at shear cracks
4. Debonding at flexural cracks



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

Following fib Bulletin

1. Anchorage failure
2. End shear failure (Concrete rip-off)
3. Debonding at shear cracks
4. Debonding at flexural cracks

Following Oehlers

Plate End debonding

(PE - debonding)

Critical Diagonal Crack debonding

(CDC - debonding)

Intermediate Crack debonding

(IC - debonding)



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

Following fib Bulletin

1. Anchorage failure
2. End shear failure (Concrete rip-off)
3. Debonding at shear cracks
4. Debonding at flexural cracks

Following Oehlers

Plate End debonding

(PE - debonding)

Critical Diagonal Crack debonding

(CDC - debonding)

Intermediate Crack debonding

(IC - debonding)



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

Following fib Bulletin

1. Anchorage failure
2. End shear failure (Concrete rip-off)
3. Debonding at shear cracks
4. Debonding at flexural cracks

Following Oehlers

Plate End debonding

(PE - debonding)

Critical Diagonal Crack debonding

(CDC - debonding)

Intermediate Crack debonding

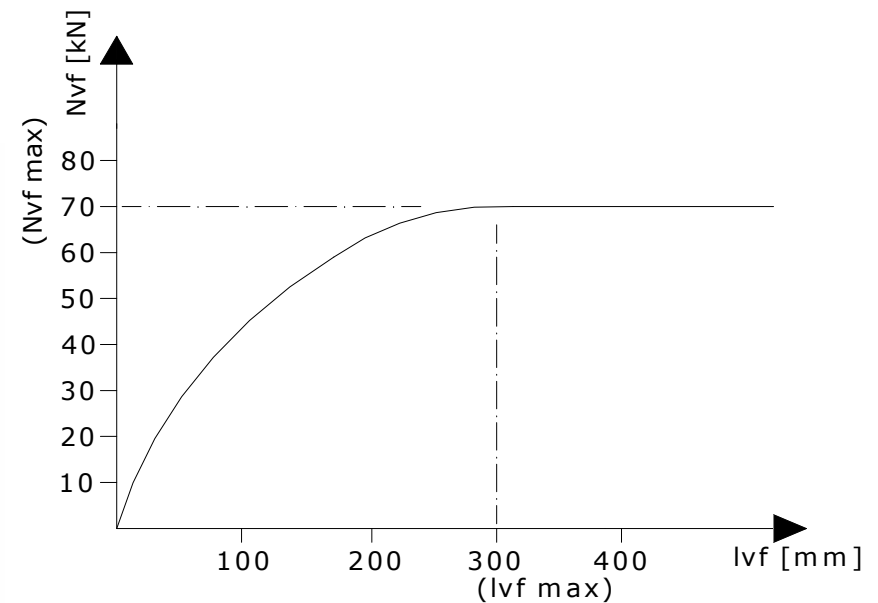
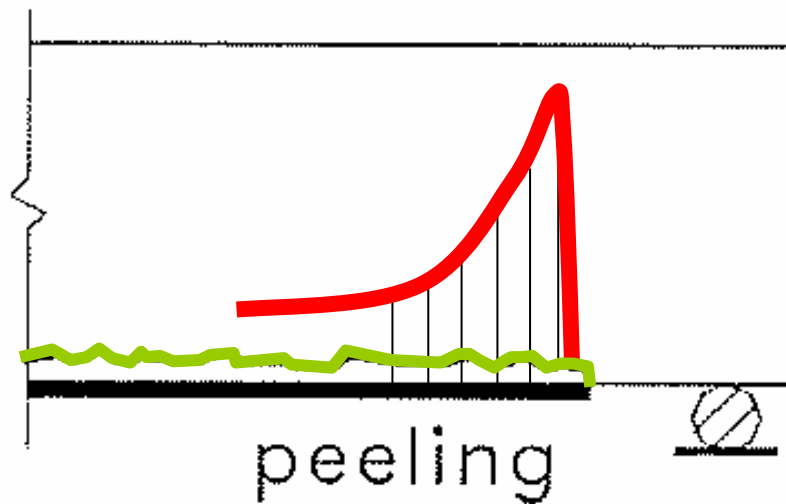
(IC - debonding)



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

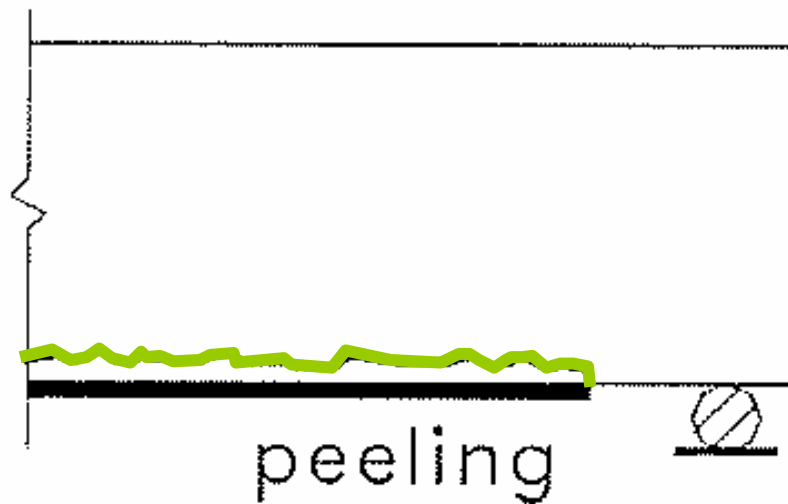
1. Anchorage length



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

1. Anchorage length



$$\frac{M_{sd}(x)}{z} = N_{fad}(x) \left(1 + \frac{A_s E_s \varepsilon_s}{A_l E_l \varepsilon_l} \right)$$

$$F_{l,max} = \alpha_H c_1 k_b k_c b_l \sqrt{f_{ctm} E_l t_l}$$

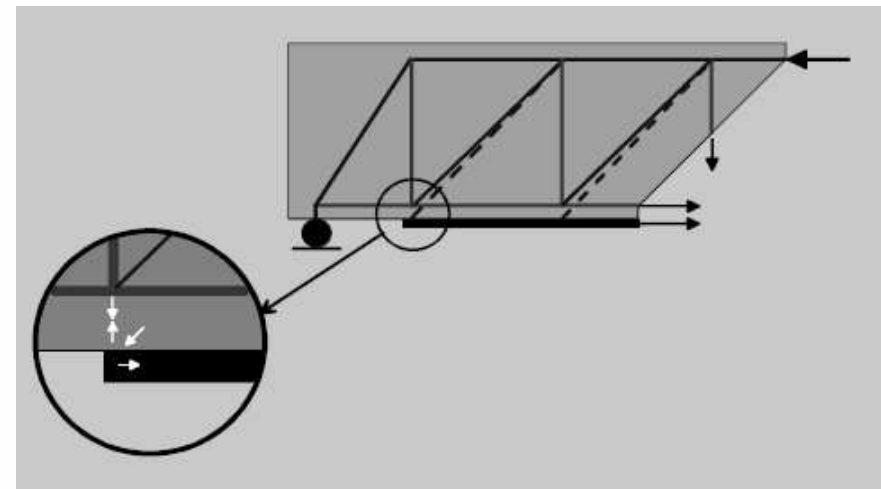
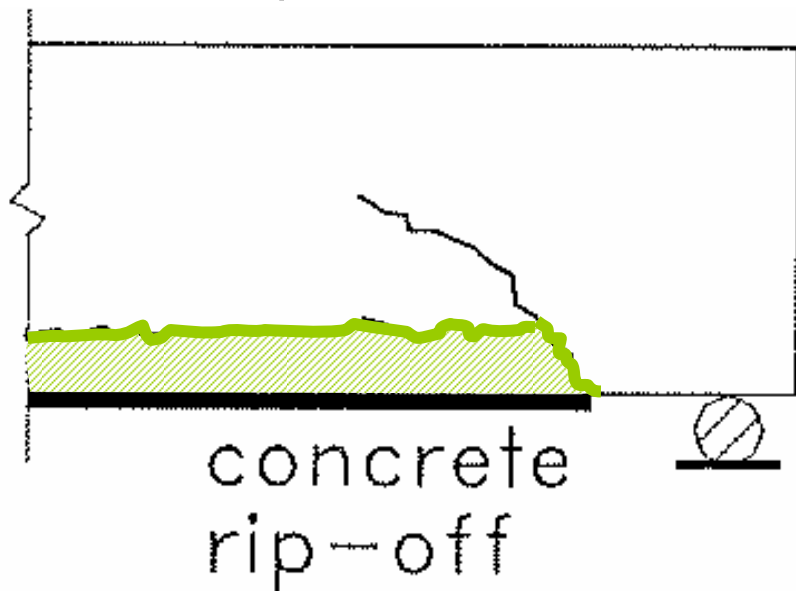
$$l_t = \sqrt{\frac{E_l t_l}{c_2 f_{ctm}}}$$



APPLICATIONS – Flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

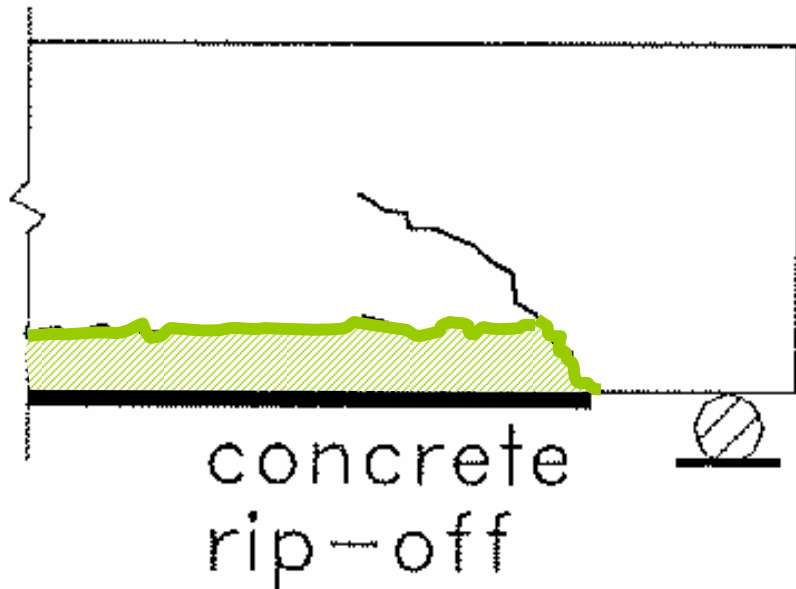
2. Concrete rip-off



APPLICATIONS – Flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

2. Concrete rip-off



$$V_{sd} \leq V_{Rd1} = \tau_{Rd1} b d$$

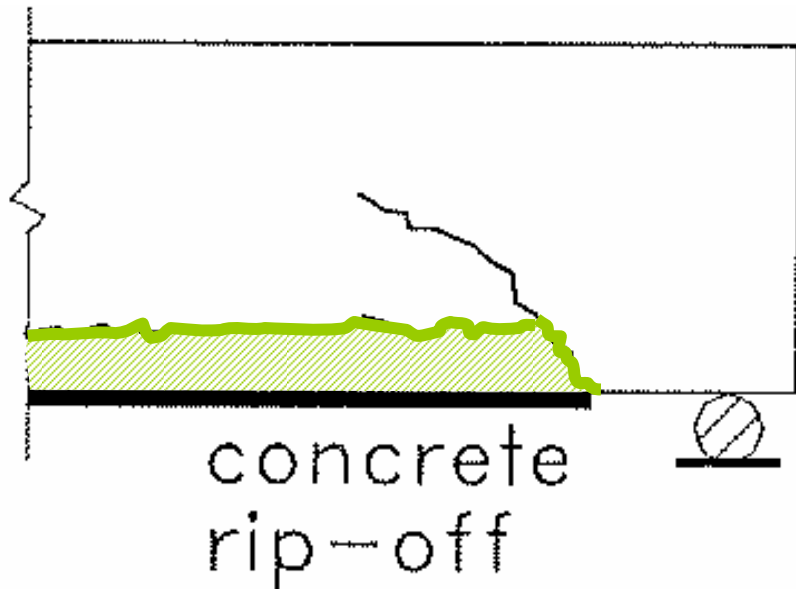
$$\tau_{Rd1} = 0,15 \left(3 \frac{d}{a_L} \right)^{1/3} \left(1 + \sqrt{\frac{200}{d}} \right) (100 \rho_s f_{ck})^{1/3}$$



APPLICATIONS – Flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

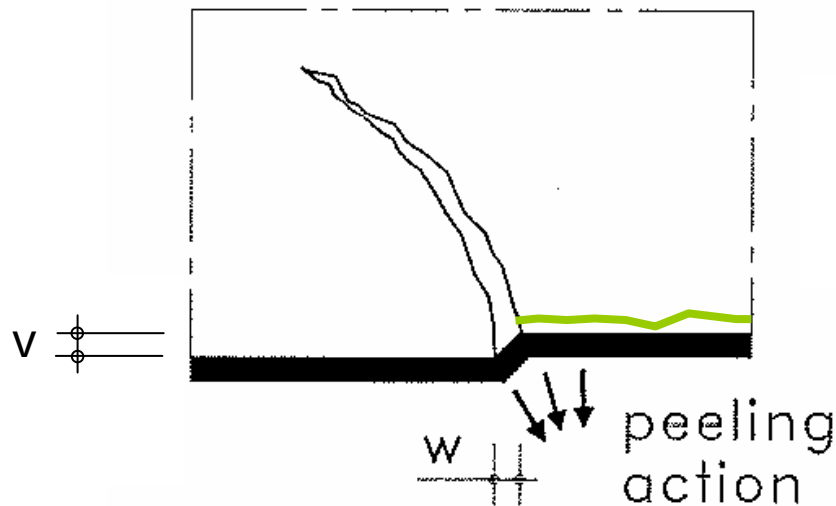
2. Concrete rip-off



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

3. Debonding at shear cracks



$$V_{Rp} = \tau_{Rp} b d$$

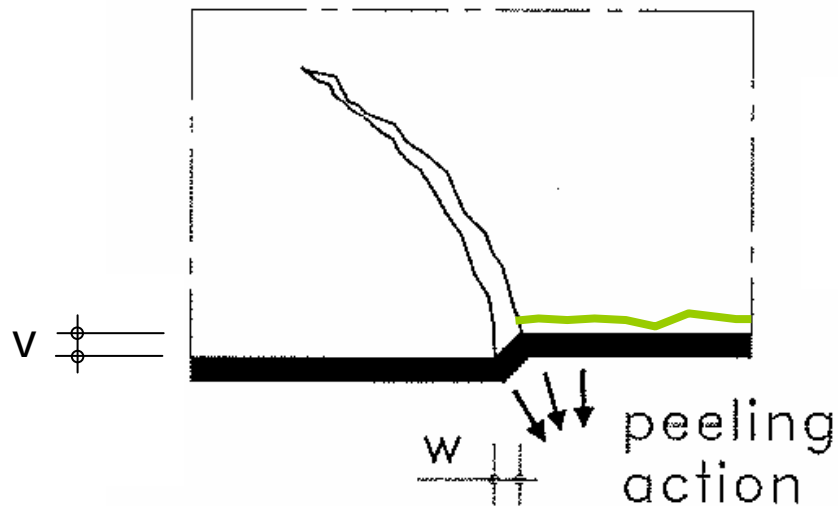
$$\tau_{Rp} = 0,38 + 151 \rho_{eq}$$



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

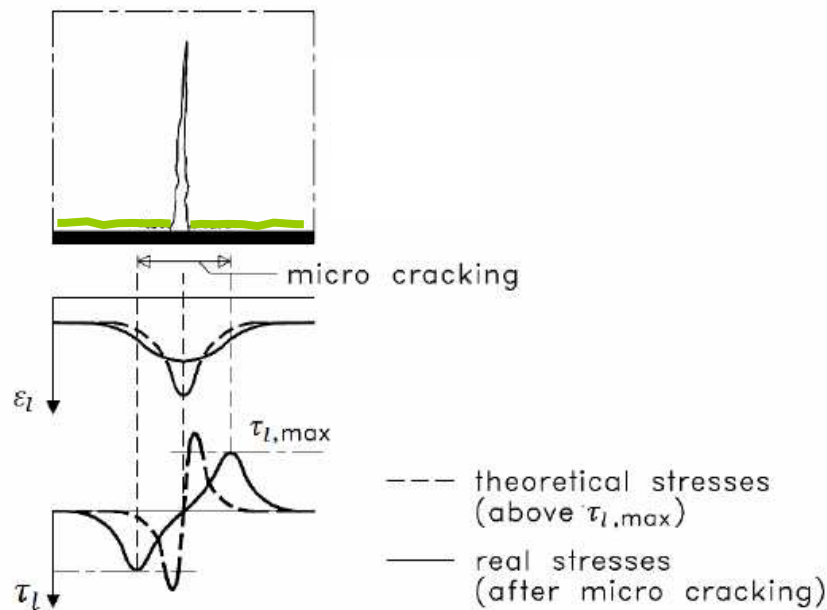
3. Debonding at shear cracks



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

4. Debonding at flexural cracks



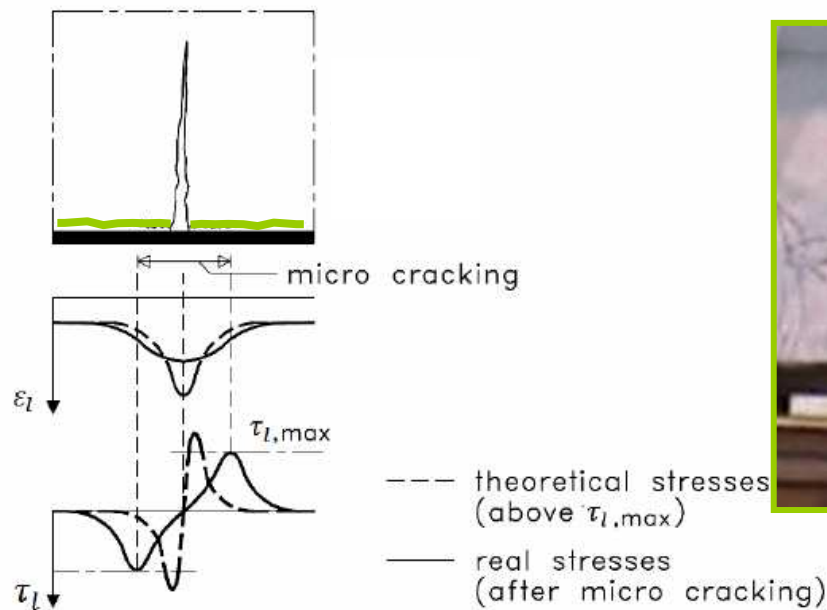
$$\tau_b = \frac{\Delta N_{fd}}{b_f \Delta x}$$



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms

4. Debonding at flexural cracks



APPLICATIONS – flexural strengthening

ULS: Ultimate limit state - Debonding mechanisms



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

SLS: Serviceability limit state

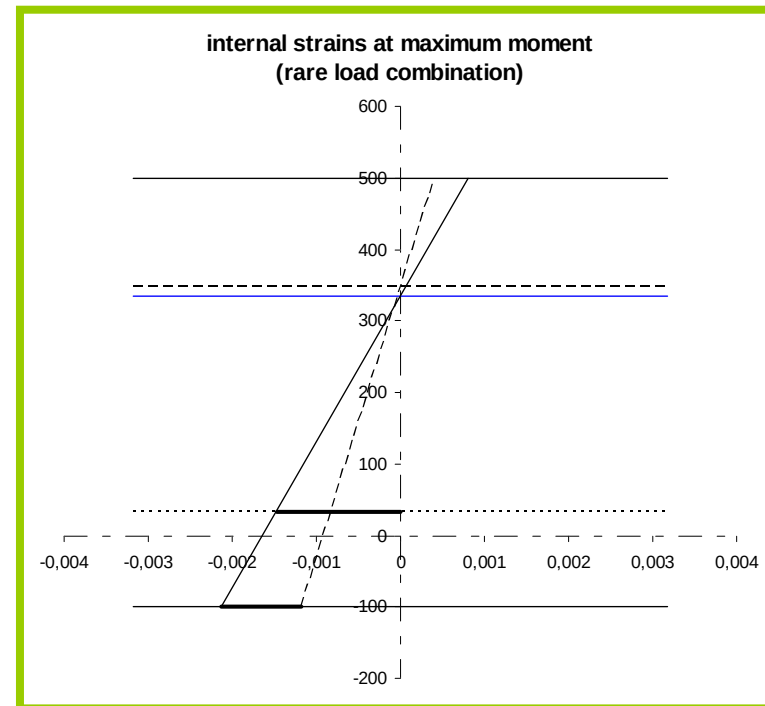
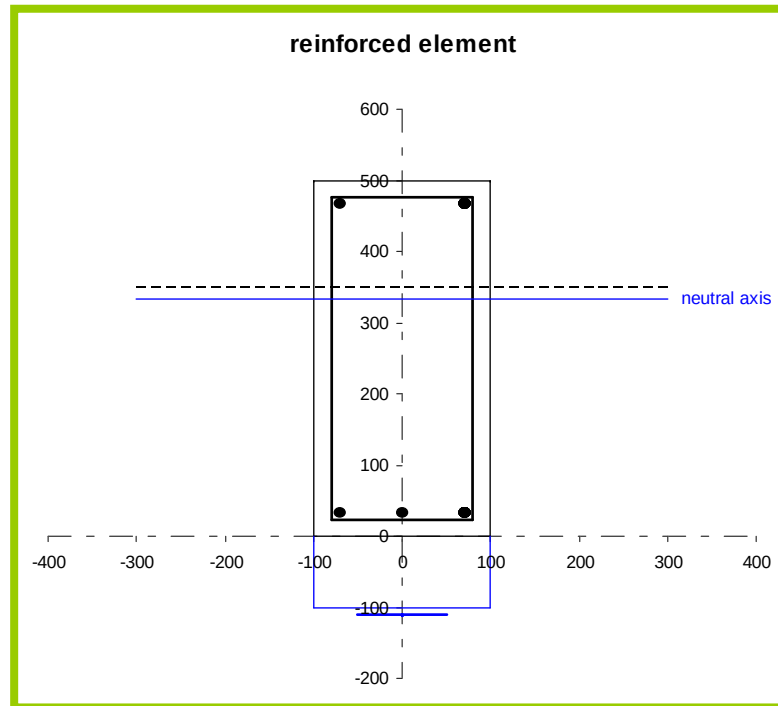
TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – flexural strengthening

SLS: Serviceability limit state



$$0,5bx_e^2 + (\alpha_s - 1)A_{s2}(x_e - d_2) = \alpha_s A_{s1}(d - x_e) + \alpha_f A_f[(h+a) - (1 + \varepsilon_0 / \varepsilon_c)x_e]$$



APPLICATIONS – flexural strengthening

SLS: Serviceability limit state

Control of the deflection:

$$a = a_1(1 - \zeta_b) + a_2 \zeta_b$$

a_1, a_2 : deflection in cracked and non-cracked state

ζ_b : tension stiffening coefficient



APPLICATIONS – flexural strengthening

SLS: Serviceability limit state

Controlle crack widths:

$$w_k = \beta s_{rm} \zeta \varepsilon_2 < 0.3 \text{ mm}$$

s_{rm} : average crack width

ε_2 : strain of tensile reinforcement



TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

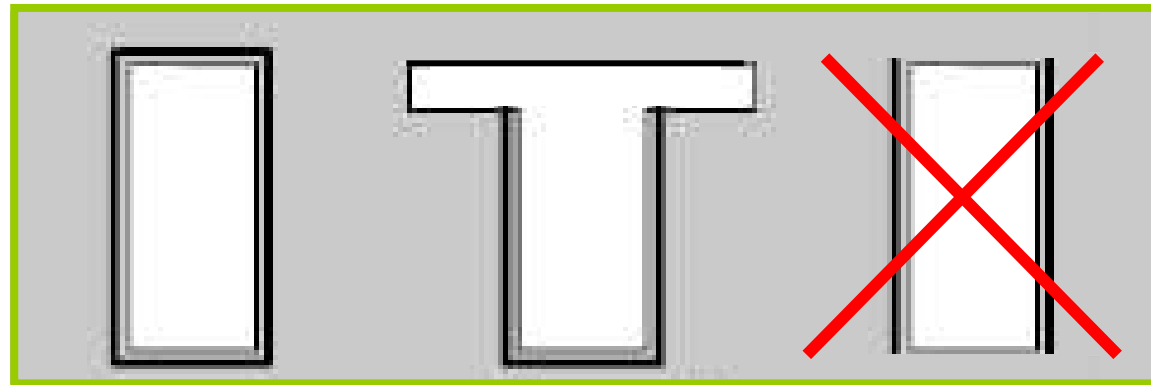
- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – shear strengthening



Complete
developed

U-shaped
reinforcement

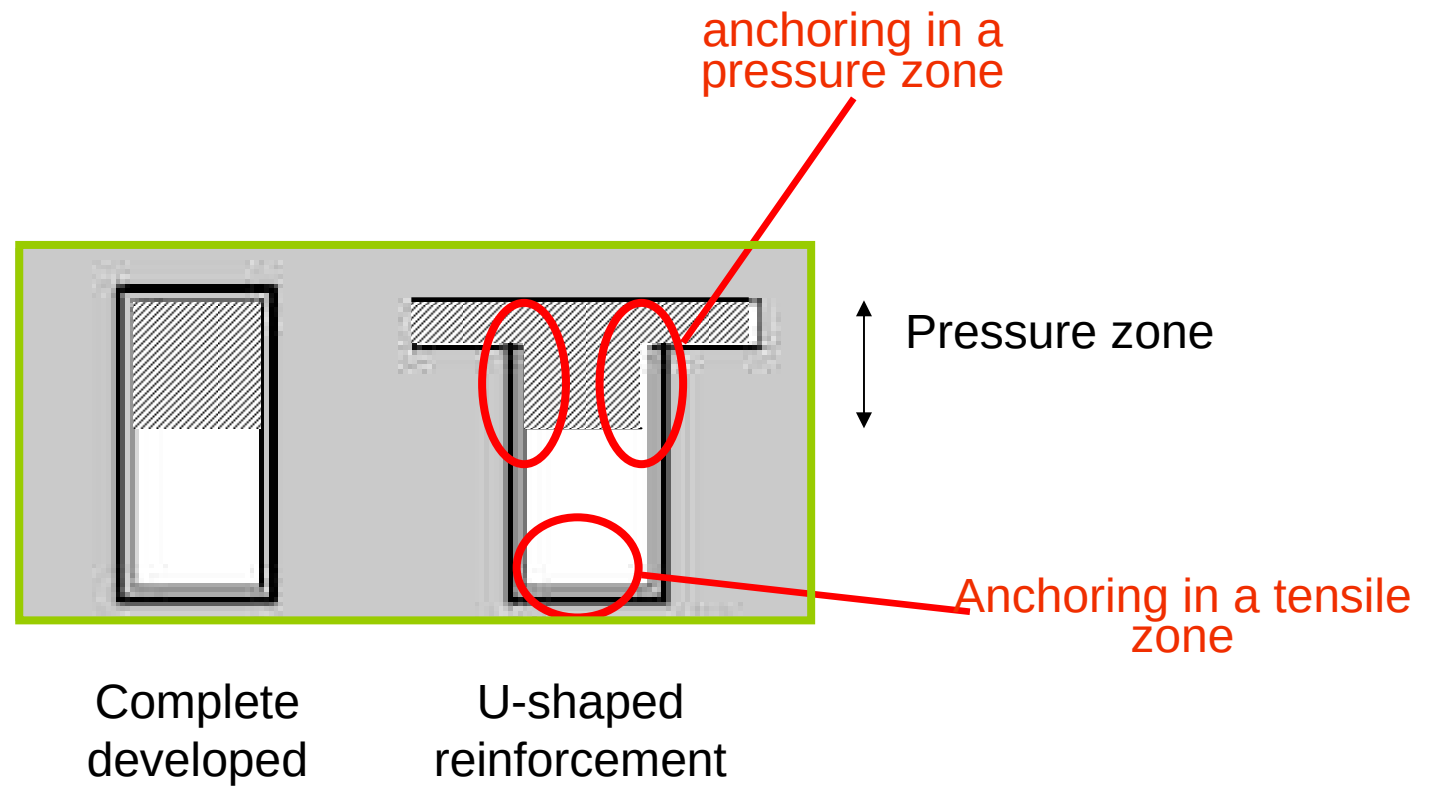
lateral
reinforcement

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen

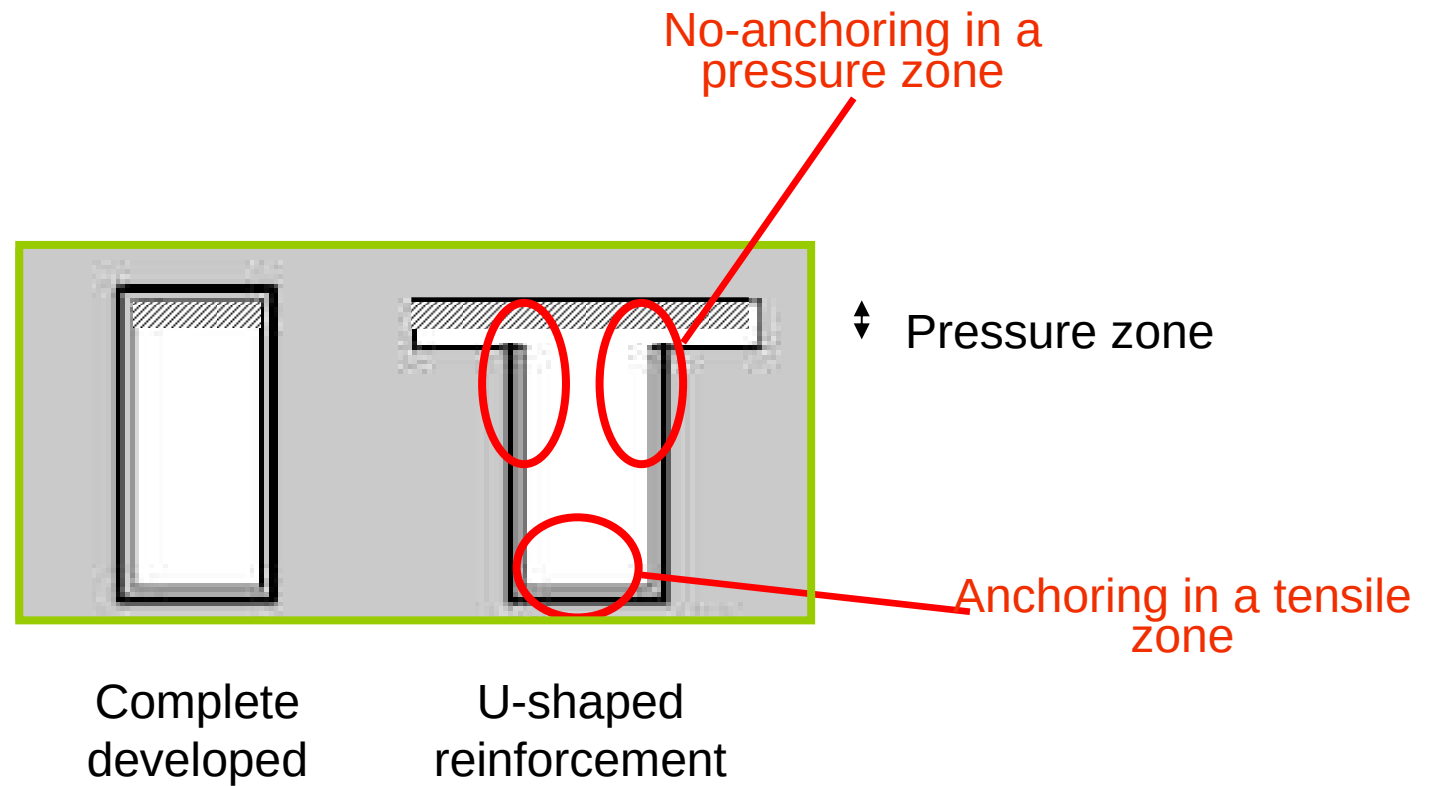


TRADECC

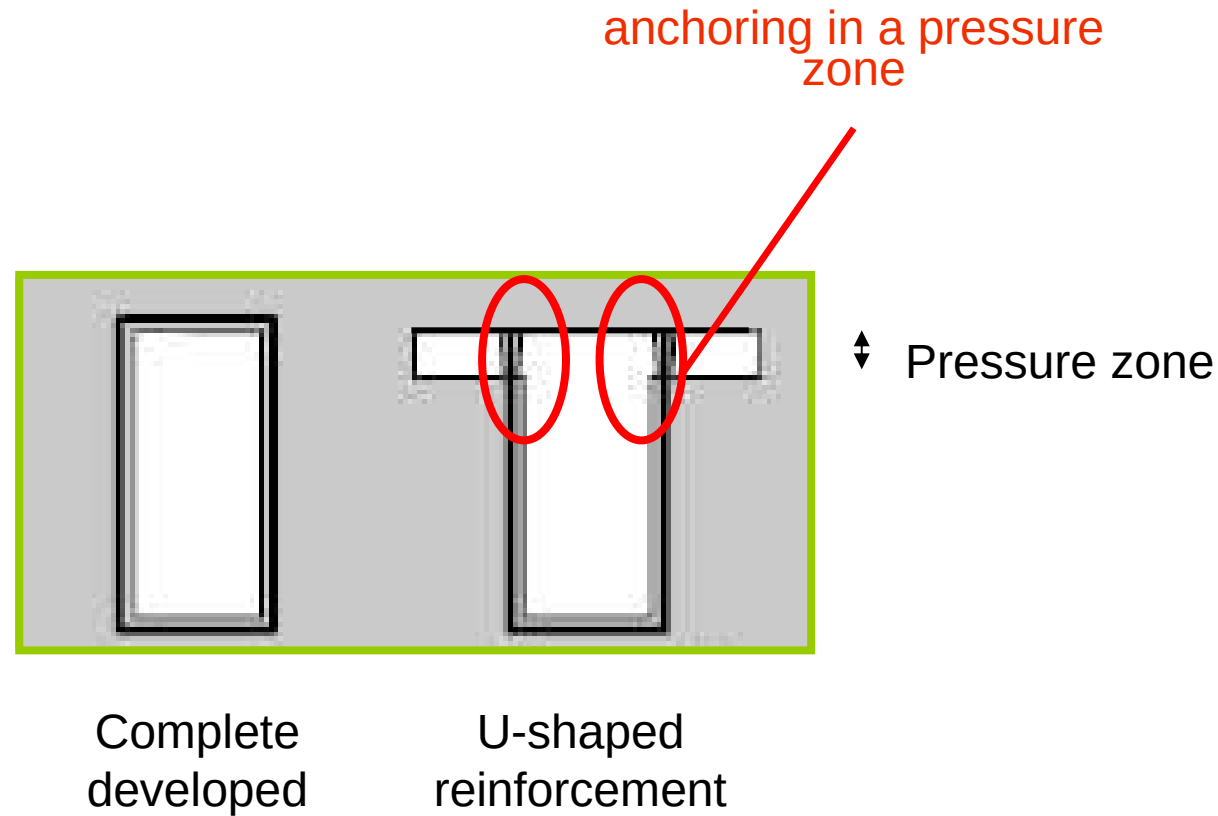
APPLICATIONS – shear strengthening



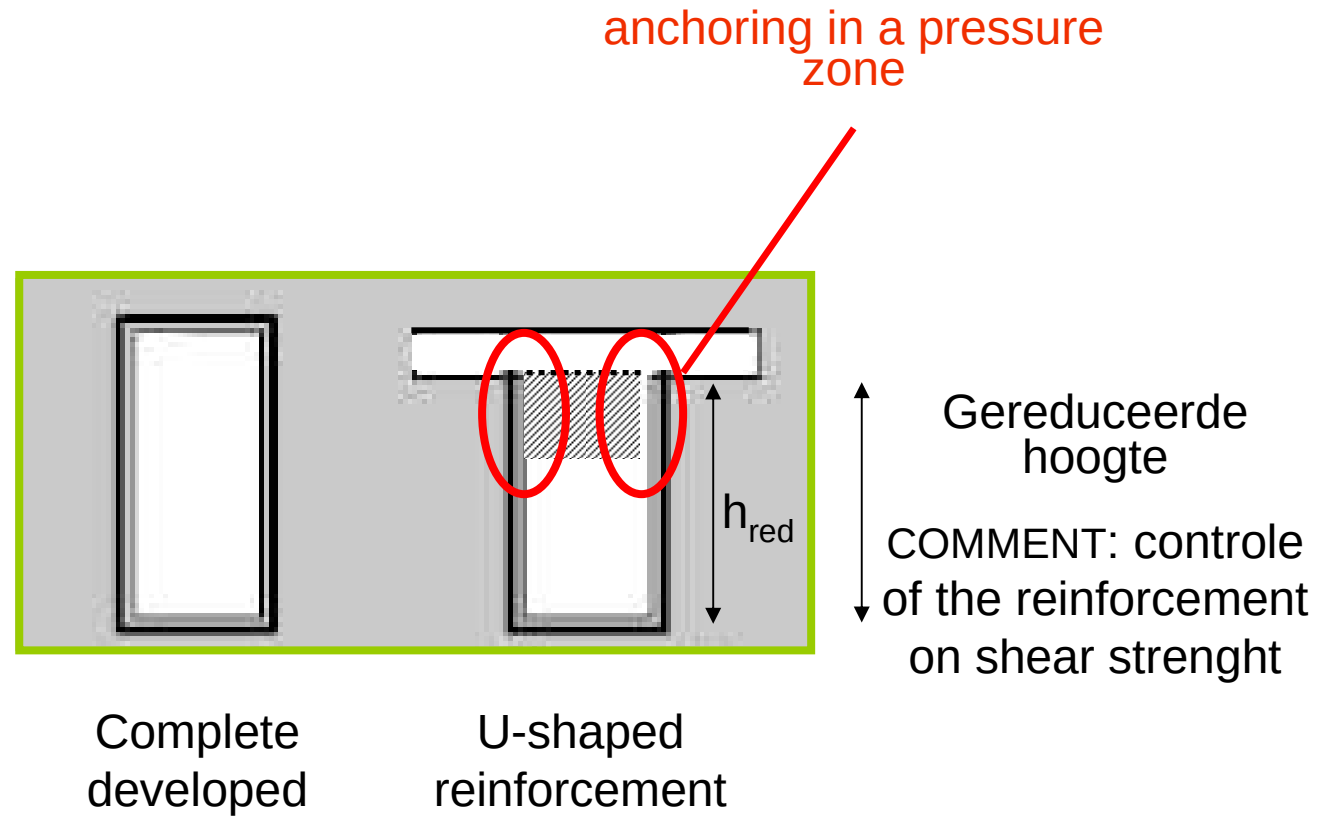
APPLICATIONS – shear strengthening



APPLICATIONS – shear strengthening



APPLICATIONS – shear strengthening



TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

APPLICATIONS – wrapping columns

check of:

- Bearing capacity (~ strength)

$$F_{Ed} = A \cdot \sigma_{cd}$$

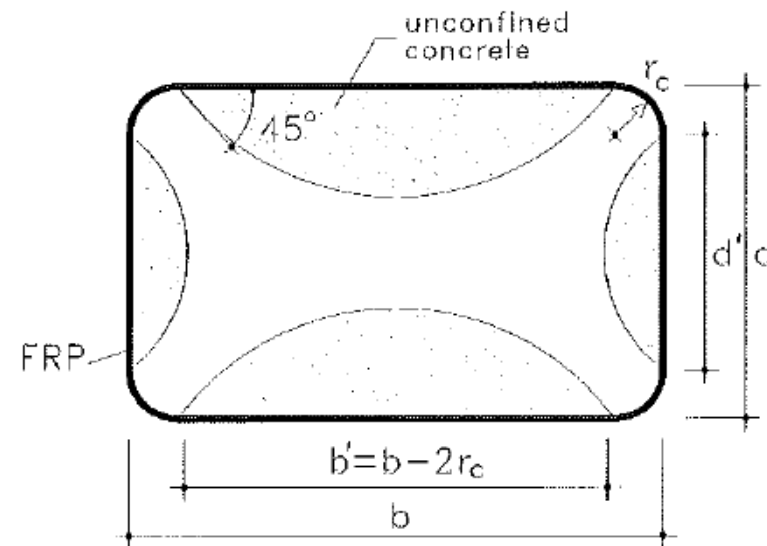
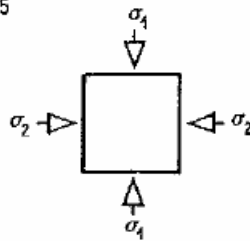
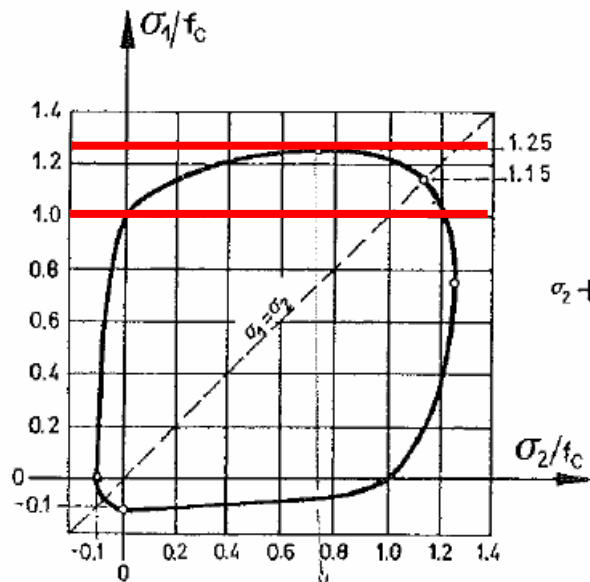


APPLICATIONS – wrapping columns

check of:

- Bearing capacity ($\sim$ strength)

$$F_{Ed} = A \cdot \sigma_{cd}$$



APPLICATIONS – wrapping columns

check of:

- Bearing capacity (~ strength)

$$F_{Ed} = A \cdot \sigma_{cd}$$

- Slenderness (~ stiffness)

To avoid buckling



TRADECC

- INTRODUCTION

- TYPE OF FIBRES

- APPLICATIONS

Flexural Strengthening

Shear Strengthening

Wrapping of Columns

- EXAMPLES

TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

BRUSSEL - Inami Riziv



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

BRUSSEL - Inami Riziv



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

BRUSSEL - Inami Riziv



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

BRUSSEL - Inami Riziv



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

BRUSSEL - Inami Riziv



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

DOORNIK - Bageci



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

DOORNIK - Bageci



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Brussel - Congrespaleis



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Brussel – Congrespaleis



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Vilvoorde – Bridge (B13) over E19



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Vilvoorde – Bridge (B13) over E19



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Vilvoorde – Bridge (B13) over E19



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC

EXAMPLES

Vilvoorde – Bridge (B13) over E19



TRADECC nv
Terbekehofdreef 50-52
2610 Wilrijk - Antwerpen



TRADECC